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PALEOMAGNETIC DATA FROM TERTIARY IGNEOUS ROCKS, NORTHEAST JALISCO, MEXICO

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New paleomagnetic measurements have been made on Tertiary volcanic rocks from northeast Jalisco, Mexico (20.7°N, 102.3°W). The pole position obtained from this study (68°N, 181°E) is consistent with two other Oligocene poles from Mexico. Mexican poles form a coherent group which differs from poles of similar ages from North America. This suggests a possible tectonic rotation of the sampling sites of Mexico with respect to "stable" North America.

1. Introduction

Paleogeographic reconstructions of Mexican regions based on paleomagnetism, although of worldwide interest, are still handicapped because only a few paleomagnetic data are available on Mexican rocks [1–8].

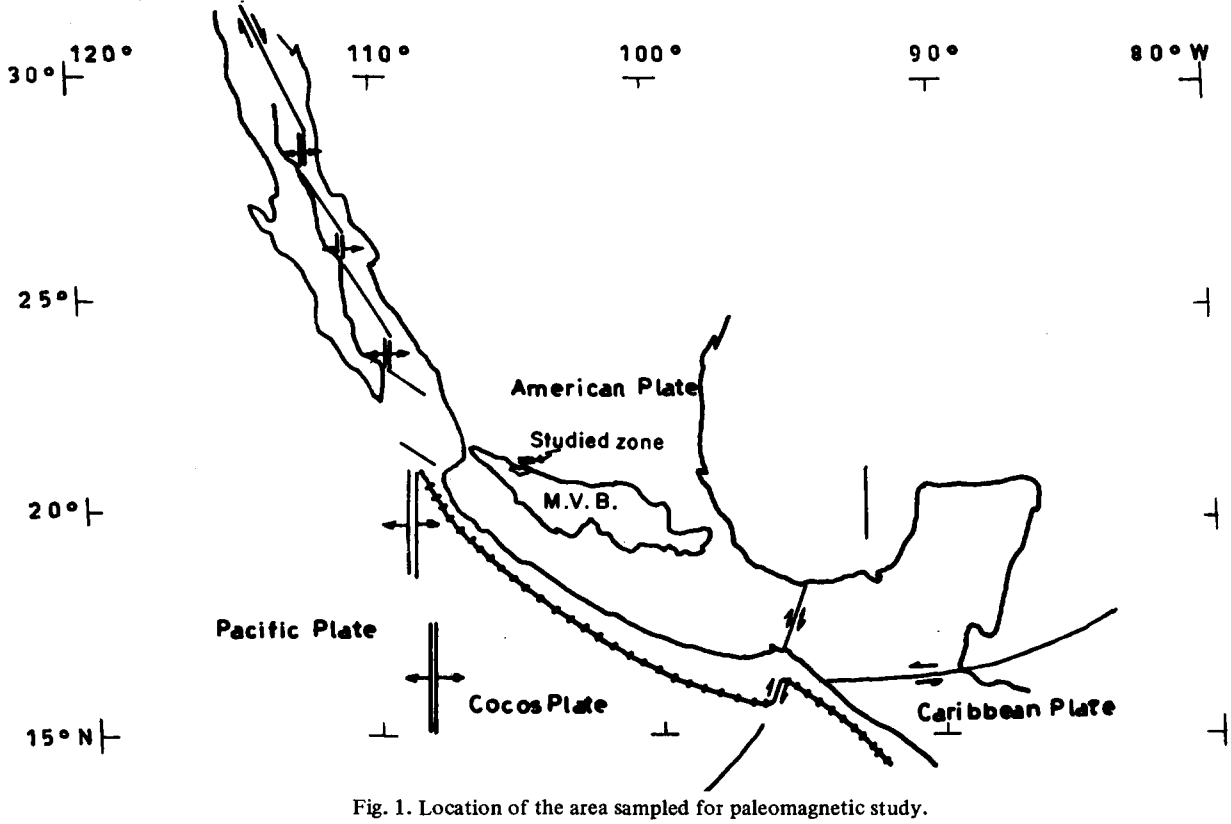


Fig. 1. Location of the area sampled for paleomagnetic study.

Earlier attempts to estimate pre-drift positions of Mexico and Central America have not been successful as is shown by considerable overlap of these areas in the reconstructions [9].

The present study reports new paleomagnetic data from an area situated north of the Mexican Volcanic Belt (MVB), near Chapala lake, Jalisco state, Mexico (Fig. 1). The extensive igneous rocks from this area have been assigned a middle Cenozoic age (Oligocene to Lower Pliocene) [10]. Location of the sampling sites is given in Table 1. Sixty-five oriented samples were collected using a core-drill and one or more specimens of length 2.54 cm were cut from each core for paleomagnetic measurements. The direction and inten-

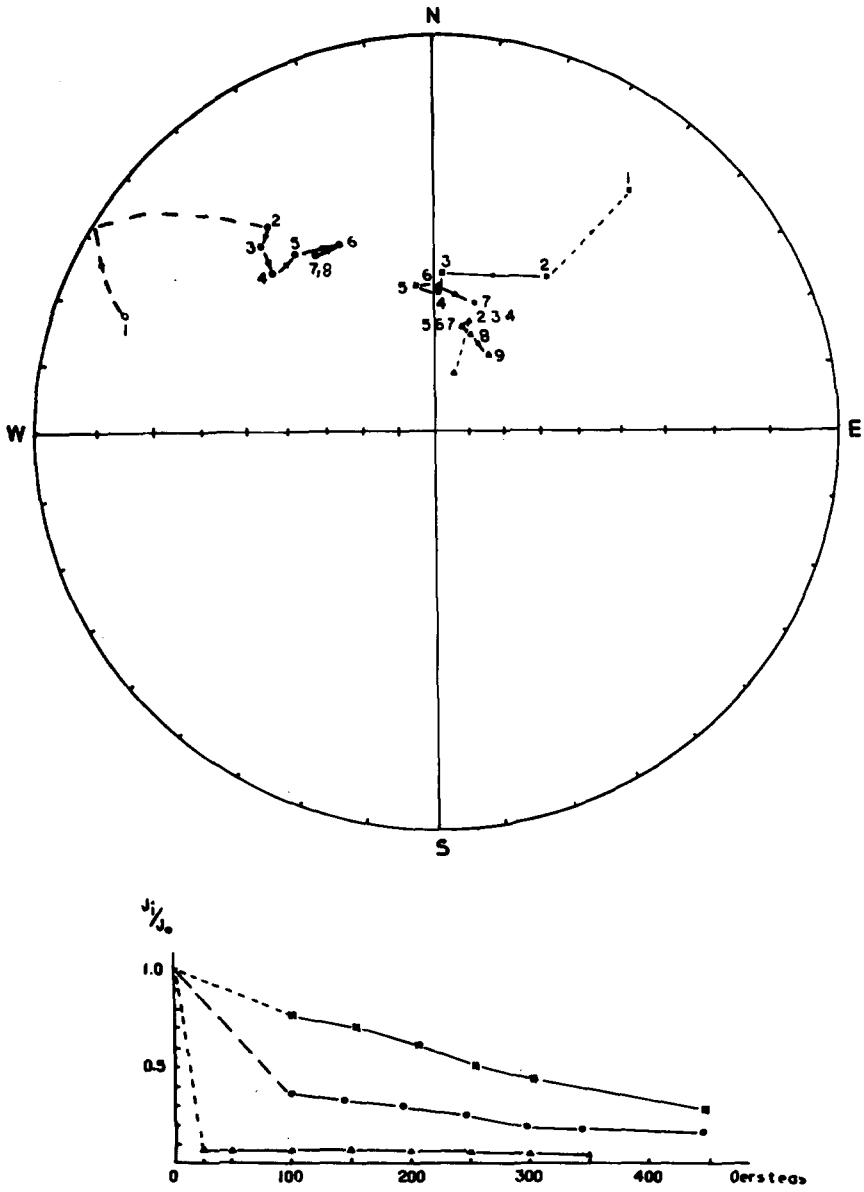


Fig. 2. Typical changes in intensity and direction of remanent magnetization: ● = site I; ■ = site II; ▲ = site V (filled symbols represent downward inclination, while open symbols indicate upward inclination).

TABLE 1

Summary of paleomagnetic data of igneous rocks from northeast Jalisco, Mexico.

Site	Location		Cleaned remanent magnetization										a.c. field	Polar coordinates			
	lat. N	long. W	N	R	D	I	K	α_{95}	δ	θ_{50}	θ_{63}	θ_{95}	(Oe)	lat. N	long. E	dp	dm
I	20°40.9′	102°12.5′	7	6.82	317.69	36.16	32.93	10.67	13.10	11.76	14.12	24.39	150	50.45°	185.40°	7.22°	12.41°
II	20°42.2′	102°14.3′	8	7.74	352.17	47.01	26.46	10.97	14.48	13.12	15.75	27.22	250	79.65°	144.16°	9.16°	14.18°
III	20°40.4′	102°09.5′	8	7.93	339.66	39.35	107.39	5.37	7.33	6.52	7.83	13.53	200	70.39°	184.94°	3.47°	5.83°
IV	20°37.3′	102°12.8′	7	6.26	323.37	40.88	8.08	22.61	26.63	23.75	28.50	47.25	250	56.02°	180.61°	16.65°	27.44°
V	20°43.1′	102°31.9′	8	7.21	342.82	23.54	8.84	19.73	25.50	22.71	27.25	47.10	200	71.52°	168.08°	11.00°	21.03°
VI	20°43.1′	102°31.9′	4	3.995	350.22	33.46	625.00	3.68	2.81	2.70	3.24	5.60	250	80.47°	179.44°	2.38°	4.19°
VII	20°41.3′	102°32.6′	6	5.21	154.65	−28.47	6.34	28.65	29.70	26.80	32.16	55.58	100	65.29°	183.85°	17.27°	31.45°
VIII	20°46.9′	102°26.1′	9	8.31	4.50	−33.56	11.57	15.80	22.61	19.84	23.81	41.16	250	50.62°	289.18°	10.25°	17.99°
Mean	20°41.9′	102°21.5′	7	6.83	335.60	36.22	35.84	10.22	12.56	11.27	13.53	23.38		68.06°	181.10°	6.92°	11.90°

N = number of samples; R = resultant vector; D = declination (degrees); I = inclination (degrees); K = precision estimate; α_{95} = semiangle of the cone of 95% confidence about the mean direction; δ = angular standard deviation of directions from the mean direction; θ_{50}, θ_{63}, θ_{95} = the quartile distance, the standard deviation and the 95% deviation, respectively, for normal distributions; dp, dm = semiaxes of the oval of confidence about the pole position at the 95% probability level.

TABLE 2

Magnetic minerals of the sampling sites

Site	Rock type	Polarity	Magnetic minerals (approximate weight (%))
I	olivine-poor basalt	N	magnetite (7)
II	olivine-poor basalt	N	magnetite (7)
III	basaltic andesite	N	magnetite (4), hematite (2)
IV	basaltic andesite	N	magnetite (5)
V	basaltic andesite	N	magnetite (6)
VI	basaltic andesite	N	magnetite (4), hematite (4)
VII	olivine andesite	R	magnetite (7)
VIII	olivine-poor basalt	N	magnetite (6)

sity of remanent magnetization of each specimen were measured using a commercial spinner magnetometer (PAR-SM2), and their stability was tested using an alternating field demagnetization apparatus [11]. Typical a.c. demagnetization curves as well as changes in the direction of remanent magnetization are shown in Fig. 2. The NRM intensities of the specimens are distributed between 1.73×10^{-6} and 1.68×10^{-3} emu/cm^3, with arithmetic mean 1.36×10^{-4} emu/cm^3. Polished thin sections were prepared from one core of each site for opaque mineralogy using a reflection microscope and the results are given in Table 2.

Mean directions of remanent magnetization were computed using Fisher statistics [12].

2. Results and discussion.

The results obtained are summarized in Table 1. Virtual geomagnetic poles for the eight sites are also given in Table 1 and plotted in Fig. 3. After cleaning, igneous rocks from site VII have stable remanent magnetization of reversed polarity and from site VIII have upward inclination, whereas those from the other sites have normal polarity. Unfortunately it is not possible to determine stratigraphic relationship of these volcanic units from topography of the sampling sites as geology of the area has not so far been studied in detail.

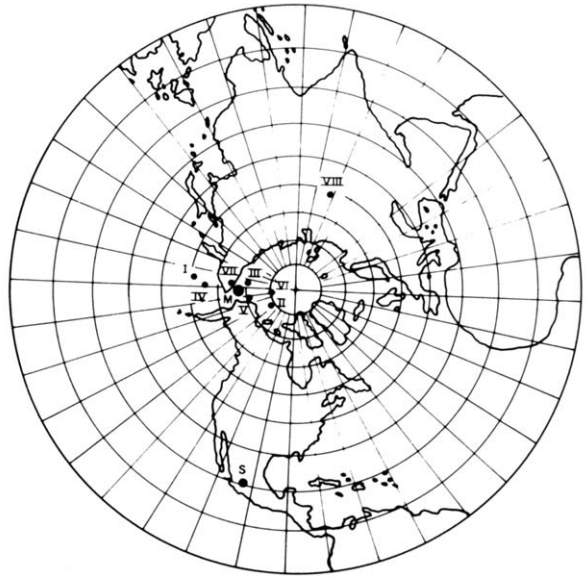


Fig. 3. Virtual geomagnetic pole positions from the eight sites, and also the mean pole position (M) from seven sites (see text).

The mean pole position obtained is 68°N, 181°E ($n = 7$, $dp = 7°$, $dm = 12°$) using mean directions of the cleaned remanent magnetizations of seven sites (site VIII was rejected). In Table 3 are summarized several pole positions from U.S.A. as well as three positions from Mexico.

The Mexican poles form a fairly well-defined group

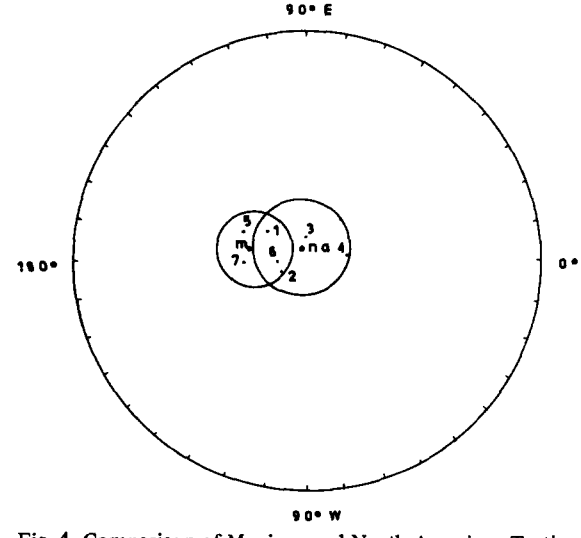


Fig. 4. Comparison of Mexican and North American Tertiary poles (m and na are the mean pole positions for Mexico and North America, respectively).

TABLE 3

Pole positions for North America

Pole No.	Location	Formation	Age (m.y.)	Pole position	Reference
1	38.5°N, 115°W	Nevada ignimbrites	To-m (30–18)	73°N, 142°E	[13]
2	37.4°N, 105°W	Spanish Peaks dyke swarm	Te-o	81°N, 211°E	[13]
3	29.3°N, 103.3°W	Buck Hill volcanic series	To (36–27)	81°N, 89°E	[13]
4	– –	SE Arizona volcanics	To (30–25)	75°N, 11°E	[2]
5	24°N, 105°W	Durango rhyolitic ignimbrites	To (28)	66.6°N, 155.6°E	[4]
6	19.5°N, 99.5°W	Valley of Mexico volcanics	To	80°N, 177°E	[14]
7	20.7°N, 102.3°W	NE Jalisco volcanics	To-p	68°N, 181°E	this work

(mean pole, 72°N, 169°E; $K = 87$; $\alpha_{95} = 13°$) which permits us to assign tentatively an Oligocene age to the rocks from Jalisco. The poles from U.S.A. (mean, 85°N, 109°E; $K = 34$; $\alpha_{95} = 16°$) do not form a well-defined group, and are distinct from the group formed by Mexican poles (Fig. 4), which possibly suggests that the sampling sites of Mexico have undergone a rotation since the rocks were magnetized.

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